

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No:- 30	
Name of Student	Arpit Chakraborty
PRN No.	25070122047
Batch	2025-29
Class	CSE A2
Academic Year & Semester	2026-27, III
Date of Submission	27/10/26
Title of Assignment:	Write a JSP application to demonstrate CRUD operations.
Practice questions **Output Screenshots Solve questions on given link and attach screen shot of leetcode with your login and o/p**	1. Create a JSP-based application to add, update, delete, and view employee details 2. Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status. Employee.jsp <%@ page import="java.sql.*" contentType="text/html;charset=UTF-8" %> <%! String safe(String s) { if (s == null) return ""; return s.replace("&", "&")

```
                .replace("<", "&lt;")
                .replace(">", "&gt;")
                .replace("\"", "&quot;")
                .replace("'", "&#x27;");
    }
%>

<%
    request.setCharacterEncoding("UTF-8");

    String url = "jdbc:mysql://localhost:3306/college_db";
    String dbUser = "javaapp";
    String dbPassword = "JavaApp123!";
    String message = "";
    String action = request.getParameter("action");

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        try (Connection con = DriverManager.getConnection(url, dbUser,
dbPassword)) {

            if ("add".equals(action)) {
                String sql = "INSERT INTO employee (emp_id, emp_name, department,
salary) VALUES (?, ?, ?, ?)";

                try (PreparedStatement ps = con.prepareStatement(sql)) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.setString(2, request.getParameter("name"));
                    ps.setString(3, request.getParameter("department"));
                    ps.setDouble(4, Double.parseDouble(request.getParameter("salary")));
                    ps.executeUpdate();
                    message = "Employee added successfully";
                }
```

```
        } else if ("update".equals(action)) {
            String sql = "UPDATE employee SET emp_name=?, department=?,
salary=? WHERE emp_id=?";

            try (PreparedStatement ps = con.prepareStatement(sql)) {
                ps.setString(1, request.getParameter("name"));
                ps.setString(2, request.getParameter("department"));
                ps.setDouble(3, Double.parseDouble(request.getParameter("salary")));
                ps.setInt(4, Integer.parseInt(request.getParameter("id")));

                int rows = ps.executeUpdate();
                message = rows == 0 ? "No employee found with that ID"
                                    : "Employee updated successfully";
            }

        } else if ("delete".equals(action)) {
            String sql = "DELETE FROM employee WHERE emp_id=?";

            try (PreparedStatement ps = con.prepareStatement(sql)) {
                ps.setInt(1, Integer.parseInt(request.getParameter("id")));

                int rows = ps.executeUpdate();
                message = rows == 0 ? "No employee found with that ID"
                                    : "Employee deleted successfully";
            }
        }
    }

  } catch (NumberFormatException e) {
    message = "ID and salary must be valid numbers";
  } catch (ClassNotFoundException e) {
    message = "MySQL JDBC driver not found";
  } catch (SQLException e) {
```

```
        message = "Database error: " + e.getMessage();
    }
%>

<!DOCTYPE html>
<html>
<head>
    <title>Employee Management</title>
</head>
<body>
    <h2>Employee Management (JSP CRUD)</h2>

    <p><b><%= safe(message) %></b></p>

    <form method="post">
        ID:
        <input type="number" name="id" required>
        <br><br>

        Name:
        <input type="text" name="name" required>
        <br><br>

        Department:
        <input type="text" name="department" required>
        <br><br>

        Salary:
        <input type="number" name="salary"
               min="0" step="0.01" required>
        <br><br>

        <button type="submit" name="action" value="add">Add</button>
        <button type="submit" name="action" value="update">Update</button>
```

```
    <button type="submit" name="action" value="delete">Delete</button>
  </form>

  <h3>All Employees</h3>

  <table border="1" cellpadding="6">
    <tr>
      <th>ID</th>
      <th>Name</th>
      <th>Department</th>
      <th>Salary</th>
    </tr>

    <%
      try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        try (Connection con = DriverManager.getConnection(url, dbUser,
dbPassword);
             Statement st = con.createStatement();
             ResultSet rs = st.executeQuery(
               "SELECT * FROM employee ORDER BY emp_id")) {

            while (rs.next()) {
    %>
    <tr>
      <td><%= rs.getInt("emp_id") %></td>
      <td><%= safe(rs.getString("emp_name")) %></td>
      <td><%= safe(rs.getString("department")) %></td>
      <td><%= rs.getDouble("salary") %></td>
    </tr>
    <%
            }
        }
```

```
            } catch (Exception e) {
        %>
        <tr>
            <td colspan="4">Unable to load employee records.</td>
        </tr>
        <%
            }
        %>
    </table>
</body>
</html>
```

Patient.jsp

```
<%@ page import="java.sql.*" contentType="text/html;charset=UTF-8" %>

<%!
    String safe(String s) {
        if (s == null) return "";
        return s.replace("&", "&amp;")
                .replace("<", "&lt;")
                .replace(">", "&gt;")
                .replace("\"", "&quot;")
                .replace("'", "&#x27;");
    }
%>

<%
    request.setCharacterEncoding("UTF-8");

    String url = "jdbc:mysql://localhost:3306/college_db";
    String dbUser = "javaapp";
    String dbPassword = "JavaApp123!";
    String message = "";
```

```
    String action = request.getParameter("action");

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");

        try (Connection con = DriverManager.getConnection(url, dbUser,
dbPassword)) {

            if ("add".equals(action)) {
                String sql = "INSERT INTO patient " +
                    "(patient_id, name, age, disease, treatment_status) " +
                    "VALUES (?, ?, ?, ?, ?)";

                try (PreparedStatement ps = con.prepareStatement(sql)) {
                    ps.setInt(1, Integer.parseInt(request.getParameter("id")));
                    ps.setString(2, request.getParameter("name"));
                    ps.setInt(3, Integer.parseInt(request.getParameter("age")));
                    ps.setString(4, request.getParameter("disease"));
                    ps.setString(5, request.getParameter("status"));

                    ps.executeUpdate();
                    message = "Patient added successfully";
                }

            } else if ("update".equals(action)) {
                String sql = "UPDATE patient SET name=?, age=?, disease=?, " +
                    "treatment_status=? WHERE patient_id=?";

                try (PreparedStatement ps = con.prepareStatement(sql)) {
                    ps.setString(1, request.getParameter("name"));
                    ps.setInt(2, Integer.parseInt(request.getParameter("age")));
                    ps.setString(3, request.getParameter("disease"));
                    ps.setString(4, request.getParameter("status"));
                    ps.setInt(5, Integer.parseInt(request.getParameter("id")));
```

```
                int rows = ps.executeUpdate();
                message = rows == 0 ? "No patient found with that ID"
                                    : "Patient updated successfully";
            }

        } else if ("delete".equals(action)) {
            String sql = "DELETE FROM patient WHERE patient_id=?";

            try (PreparedStatement ps = con.prepareStatement(sql)) {
                ps.setInt(1, Integer.parseInt(request.getParameter("id")));

                int rows = ps.executeUpdate();
                message = rows == 0 ? "No patient found with that ID"
                                    : "Patient deleted successfully";
            }
        }
    }

  } catch (NumberFormatException e) {
      message = "Patient ID and age must be valid numbers";
  } catch (ClassNotFoundException e) {
      message = "MySQL JDBC driver not found";
  } catch (SQLException e) {
      message = "Database error: " + e.getMessage();
  }
%>

<!DOCTYPE html>
<html>
<head>
  <title>Patient Records</title>
</head>
<body>
```

```
<h2>Patient Records (JSP CRUD)</h2>

<p><b><%= safe(message) %></b></p>

<form method="post">
  Patient ID:
  <input type="number" name="id" required>
  <br><br>

  Name:
  <input type="text" name="name" required>
  <br><br>

  Age:
  <input type="number" name="age" min="0" required>
  <br><br>

  Disease:
  <input type="text" name="disease" required>
  <br><br>

  Treatment Status:
  <select name="status" required>
    <option value="Admitted">Admitted</option>
    <option value="Under Treatment">Under Treatment</option>
    <option value="Recovered">Recovered</option>
    <option value="Discharged">Discharged</option>
  </select>
  <br><br>

  <button type="submit" name="action" value="add">Add</button>
  <button type="submit" name="action" value="update">Update</button>
  <button type="submit" name="action" value="delete">Delete</button>
</form>
```

```
<h3>All Patients</h3>

<table border="1" cellpadding="6">
    <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Age</th>
        <th>Disease</th>
        <th>Treatment Status</th>
    </tr>

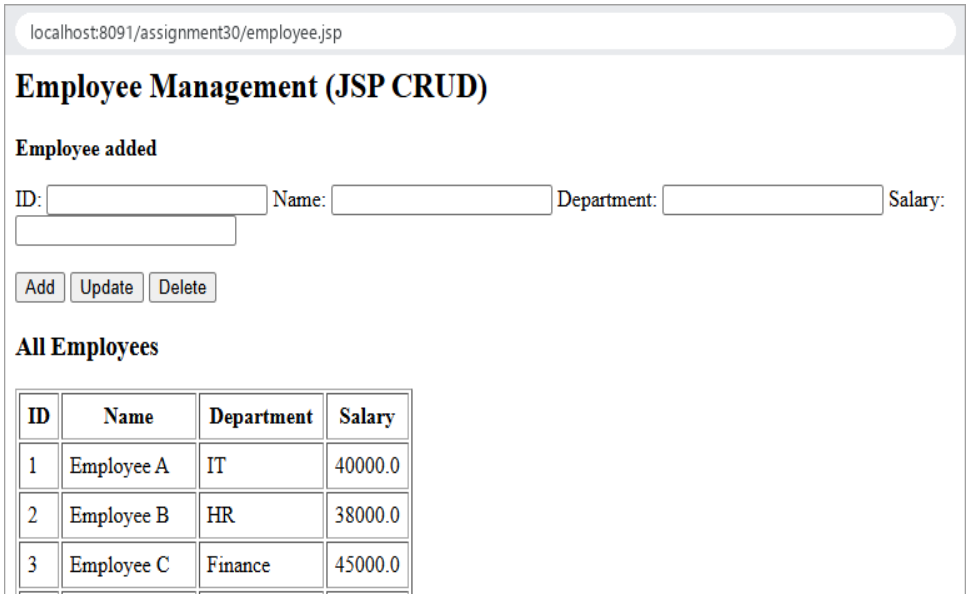
    <%
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

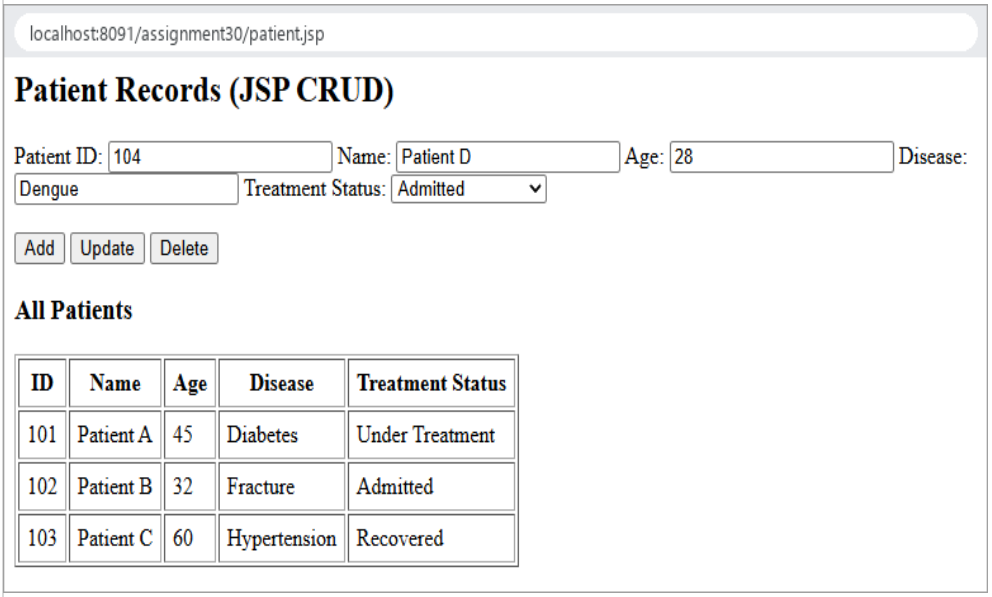
            try (Connection con = DriverManager.getConnection(url, dbUser,
dbPassword);
                 Statement st = con.createStatement();
                 ResultSet rs = st.executeQuery(
                    "SELECT * FROM patient ORDER BY patient_id")) {

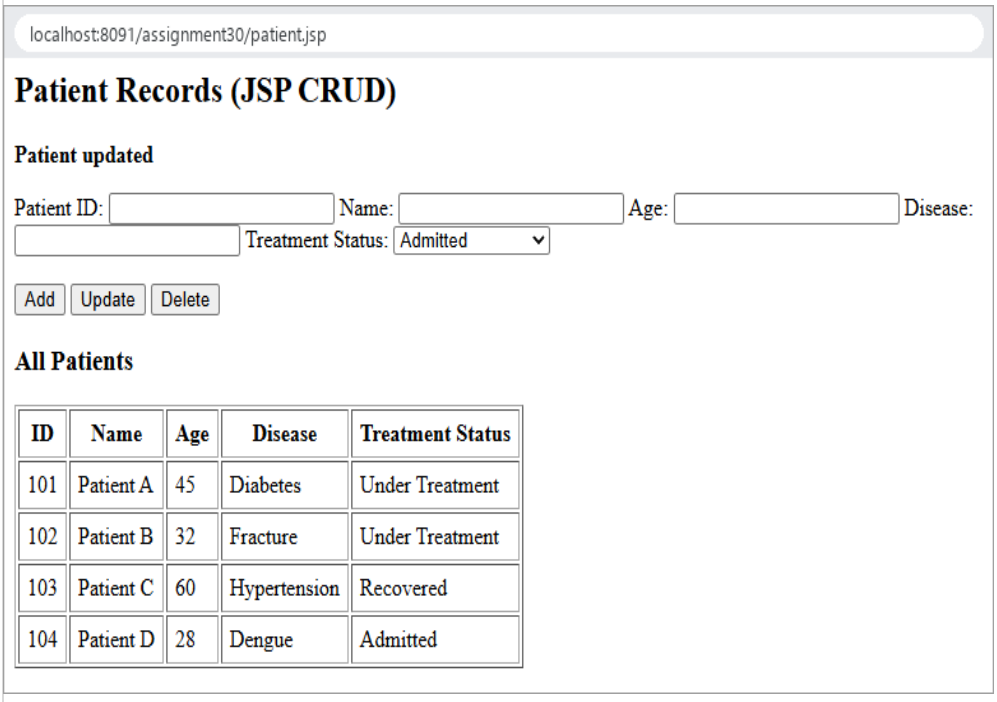
                while (rs.next()) {
    %>
    <tr>
        <td><%= rs.getInt("patient_id") %></td>
        <td><%= safe(rs.getString("name")) %></td>
        <td><%= rs.getInt("age") %></td>
        <td><%= safe(rs.getString("disease")) %></td>
        <td><%= safe(rs.getString("treatment_status")) %></td>
    </tr>
    <%
                }
            }
```

```
            } catch (Exception e) {
        %>
        <tr>
            <td colspan="5">Unable to load patient records.</td>
        </tr>
        <%
            }
        %>
    </table>
</body>
</html>
```

Output Screenshots (if applicable)







Conclusion	The Employee Management System was successfully developed using JSP, JDBC, and MySQL to perform CRUD operations, including adding, viewing, updating, and deleting employee records. Similarly, the Patient Records Management System was implemented to manage patient details such as patient ID, name, age, disease, and treatment status. Both applications demonstrate effective database connectivity and record management using JSP and MySQL.